



SPF IMMUNOASSAY ANALYZER (AU20V)

Communication specifications

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Change history

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A	11 June 2025	Sakauchi	1.0	Equipment name amended to Product name: AU20V
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1 Introduction

1.1 Positioning of this document

This specification describes the system configuration and operating procedure processing flow for connecting the SPF IMMUNOASSAY ANALYZER (AU20V) to external communication device.

2 Application

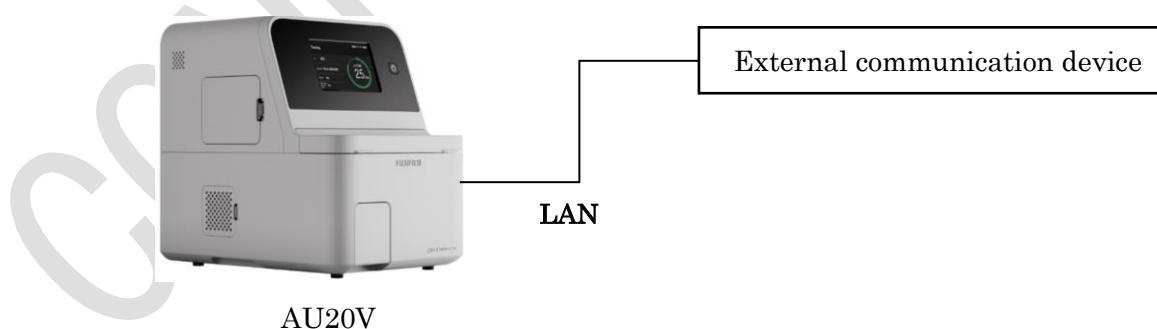
This specification applies to IF connecting SPF IMMUNOASSAY ANALYZER (AU20V) and external communication device.

3 Purpose of this document

The system specified in this specification simultaneously requests information from external communication devices from the measuring device side and displays the request details (sample number, patient name, test items, etc.) on the screen or prints them out to the user, thereby achieving the following points.

- Reduction of sample mix-ups due to sample number input errors, etc.
- Reduced operating costs through elimination of worksheets.

4 System configuration



5 Interface specifications

5.1 Overview

5.1.1 Type 1 (bi-directional communication for measurement information query)

AU20V queries the external communication device for measurement information (patient name, measurement parameters, etc.) and sends the measurement results after the measurement is completed.

When selecting this communication method, select the communication type to 'Type 1' by the host connection settings in administrator mode.

5.1.2 Type 2 (unidirectional communication for measurement information query)

This is unidirectional communication that simply transmits measurement results, but the protocol is the same as the Type 1 communication specification above.

When selecting this communication method, select the communication type to 'Type 2' by the host connection settings in administrator mode.

5.1.3 Type 3 (simple transmission of measurement results).

The same protocol as the FDC3500 that simply transmits measurement results.

When selecting this communication method, select the communication type to 'Type 3' by the host connection settings in administrator mode.

5.1.4 Type 1 + reference interval range

This communication type is added a reference interval range to Type 1 communication protocol.

When selecting this communication method, select the communication type to 'Type 1 + reference interval range' by the host connection settings in administrator mode.

5.1.5 Type 2 + reference interval range

This communication type is added a reference interval range to Type 2 communication protocol.

When selecting this communication method, select the communication type to 'Type 2 + reference interval range' by the host connection settings in administrator mode.

5.1.1 Type 3 + reference interval range

This communication type is added the animal species, outside of reference interval range and reference interval range to Type3 communication protocol.

When selecting this communication method, select the communication type to 'Type 3 + reference interval range ' by the host connection settings in administrator mode.

5.2 Communication specifications

5.2.1 LAN

Connect AU20V as a client and the external communication device side as a server.

Only IPv4 is supported, and IPv6 is not supported.

Set the IP address and port number of the external communication device in the IP address of the host connection setting on AU20V.

Socket connection shall be made to the IP address and port of the external communication AU20V that is held during host communication when using LAN.

5.3 Control specifications

5.3.1 Connection timeout monitoring

Condition	When a socket connection has not been connected for 10 seconds.
Action	The request is cleared. AU20V displays an error.

5.3.2 Response reception timeout monitoring

Condition	If no response is received within 5 seconds after issuing a processing request command.
Action	Clear the request in question. AU20V displays an error.

5.3.3 Retransmission

No retransmission

6 Message format

6.1 Message structure

6.1.1 Text structure

[STX(02H)] Command classification, parameter 1, parameter 2,[ETX(03H)][BCC].
Parameters are added as required.

The delimitation between command categories and parameters and between parameters shall be ','.

When multiple blocks are concatenated, [ETB(17H)] shall be added at the end of the block.

Within parameters, ',' and '@' shall not be used. Character codes that can be used are 20H to 7EH and A1H to DFH.

Example)

STX	Command classification	,	Parameter 1	,	Parameter 2	ETX	BCC
-----	------------------------	---	-------------	---	-------------	-----	-----

Example)

STX	Command classification	,	Parameter 1-1	,	Parameter 1-2	ETB
-----	------------------------	---	---------------	---	---------------	-----

Parameter 2-1	,	Parameter 2-2	ETX	BCC
---------------	---	---------------	-----	-----

6.1.2 BCC

The BCC calculation range shall be from the next command segment of STX to ETX.
The calculation method shall be exclusive logical OR.

Example)

STX	Command segment	,	Parameter 1	,	Parameter 2	ETX	BCC
-----	-----------------	---	-------------	---	-------------	-----	-----

Scope of calculation

6.2 Classification of telegrams

6.2.1 List of processing categories

Request/notification (Analyzer → External communication device)		Response (External communication device → Analyzer)	
Processing category	Contents	Processing category	Description
X / Y	Order index query request for AU20V	X / Y	Order index query response for AU20V
S	Measurement start notification		
R / T	Measurement result notification		
E	Error information notification		

6.2.2 Definition

Name	Contents
Processing category	Processing category represented by a single alphabetic character
Sample number※1	Number assigned to each sample by the external communication device.
Patient ID	Unique number for each patient (e.g. medical record No.)
Patient name	Patient name
Sample type	W, P, U sample type
Number of sample measurement	Number of items requested per sample
Measurement item	Name of requested measurement item
Order index	Set of sample number, patient ID, patient name, etc.
Animal type	Number assigned to each animal type
Reception number	Number for each patient (unique number for one day)

※1 Sample number: If the sample number is taken by an external communication device, it should be "year, month and day" + additional number yyymmdd**

(ex. If the sample was measured offline at 122001, only the supplementary number should be used and the month and date should not be included.

7 Details of message

7.1 Order index query request for AU20V

Used when an order query is made from the Measuring equipment (SPF IMMUNOASSAY ANALYZER) to an external communication device.

Item	Character	Size	Contents
Header	STX	1	STX:02h
Processing category	'X' 'Y'	1	Order index query command ('X': 58h) / ('Y': 59h) X: Communication type 1 Y: Communication type 1 + reference interval range
Delimited by	','	1	
Sample number	Number	13	Max. 13 characters
Separator	','	1	
Patient ID	Alphanumeric	13	Max. 13 characters
Separator	','	1	
Patient name	Alphanumeric	13	Max. 13 characters
Separator	','	1	
Number of requests	Number	2	Maximum number of indices to request 1 to 99 (max. 2 characters)
Delimiter	ETX	1	ETX: 03h
BCC.			

※The sample number shall be the first number for retrieving the order.

Blank is also acceptable.

※The search direction shall be in the order of the order. However, orders that have already been notified of the start of measurement shall be placed last.

※The results obtained from the search are returned in the order index query response.

※The sample number, patient ID and patient name are used as search keys.

※If there is data for more than one term, the order of priority shall be
Sample No. -> Patient ID -> Patient name, in that order.

※The sample number, patient ID and patient name should have a maximum of 13 characters.

Example 1: Requesting order index information for five samples without sample number to start the search.

<Transmission data>															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	X	,	,	,	,	5	ETX	BCC						

□ = space

Example 2: Requesting order index information for five samples from sample number 061201.

Sample No.				061201											
Patient ID				12345ABCD											
Patient name				Tarou Fuji											
<Transmission data>															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	X	,	0	6	1	2	0	1	,	1	2	3	4	5
2	A	B	C	D	,	T	a	r	o	u	□	F	u	j	i
	,	5	ETX	BCC											
□= space															

7.2 Order index query response for AU20V

7.2.1 Type 1

Used to notify measurement information from an external communication device to a Measuring equipment when set to Type 1.

Item	Character	Size	Contents
Header	STX	1	STX: 02h
Processing category	'X'	1	Order index query response command ('X': 58h)
Separator	','	1	
Index number	Number	2	Number of indexes to be appended afterwards (max. 2 characters)
Separator	','	1	
Sample number 1	Number	13	Max. 13 characters
Separator	','	1	
Patient ID 1	Alphanumeric	13	Max. 13 characters
Separator	','	1	
Patient name 1	Alphanumeric	13	Max. 13 characters
Separator	','	1	
Animal type 1	Number	2	0 - 99 (max. 2 characters)
Separator	','	1	
Gender 1	Number	1	0: Male 1: Female 9: Unsettled
Separator	','	1	
Age 1	Number	3	999: Unsettled (max. 3 characters)
Separator	','	1	
Number of measurements	Number	2	0 - 5 items = number of measurement items (2 characters fixed (0 padding): '00' ~ '05')
Separator	','	1	
Measurement item 1	English	6	Max. 6 characters
Separator	','	1	
Measurement item 2	English	6	Max. 6 characters
Separator	','	1	
Measurement item 3	English	6	Max. 6 characters
Separator	','	1	
Measurement	English	6	Max. 6 characters

item 4			
Separator	,	1	
Measurement item 5	English	6	Max. 6 characters
Block Separator	ETB	1	
...			
Sample number n	Number	13	Max. 13 characters
Separator	,	1	
Patient ID n	Alphanumeric	13	Max. 13 characters
Separator	,	1	
Patient name n	Alphanumeric	13	Max. 13 characters
Separator	,	1	
Animal type n	Number	2	0 - 99 (max. 2 characters)
Separator	,	1	
Gender n	Number	1	0: male 1: female 9: unspecified
Separator	,	1	
Age n	Number	3	999: not set (max. 3 characters)
Separator	,	1	
Number of measurements	Number	2	0 - 5 items = number of measurement items (2 characters fixed (0 padding): '00'~ '05')
Separator	,	1	
Measurement item 1	English	6	Max. 6 characters
Separator	,	1	
Measurement item 2	English	6	Max. 6 characters
Separator	,	1	
Measurement item 3	English	6	Max. 6 characters
Separator	,	1	
Measurement item 4	English	6	Max. 6 characters
Separator	,	1	
Measurement item 5	English	6	Max. 6 characters
Delimiter	ETX	1	ETX: 03h
BCC.			

※The sample number, the number of sheets to be measured and the number of indexes are mandatory.

※If there is no order, the number of indexes shall be set to 0. However, the sample number data added to the query request shall be used for sample number 1.

※If there is an order, the sample number is mandatory and the patient ID and Patient name must have data on either of them.

※If the sample number is empty, or if the sample number exists but both patient ID and patient name are empty, the field is invalid.

※The number of measurement sheets indicates the number of measurement items, up to a maximum of five items.

※If the number of measurement sheets is 0, it is interpreted as no order.

※If the number of measurement items is too large for the number of sheets measured, the fields after the number of sheets measured are ignored.

※If the number of measurement items is small in relation to the number of sheets, no error is made and processing continues.

※The maximum number of measurement item fields is five.

※If the number of index fields is large in relation to the number of indexes, the subsequent fields are ignored.

※If the number of index fields is small in relation to the number of indexes, no error is made and processing continues.

※The block delimiter (ETB) in the last index field must not be added.

Example: Replying to two requests.

Information of the first sample

Sample No. 2006061201
 Patient ID ABCDEFGHIJKLM
 Patient Name Taro Fuji
 Animal type 2: Cat
 Gender 1: Female
 Age 3 years old
 Number of measurements 3
 Measurement item 1 v-TSH
 Measurement item 2 v-T4
 Measurement item 3 v-CORT

Information of the second sample

Sample No. 2006061202
 Patient ID 12345ABCD
 Patient Name Lucy Heska
 Animal type 1: Dog
 Gender 0: Male
 Age 1 year old
 Number of measurements 1
 Measurement item 1 v-TSH

<Transmission data>.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	X	,	2	,	2	0	0	6	0	6	1	2	0	1
2	,	A	B	C	D	E	F	G	H	I	J	K	L	M	,
3	T	A	r	o o	□	F	u	J	i	,,	2	,	1	,	3
4	,	3	,	v	-v	T	S	H	,	v	-	T	4	,	v
5	-	C	O	R	T	TB	2	0	0	6	0	6	1	2	0
6	2	,	1	2	3	4	5	A	B	C	D	,	L	u	c
7	y	□	H	e	s	k	A	,	1	,	0	,	1	,	1
8	,	v	-	T	S	H	ETX	BCC							

□ = Space

7.2.2 Type 1 + reference interval range

Used to notify measurement information from an external communication device to the Measuring equipment when set to Type 1 + reference interval range.

※Differences from Type 1 are marked in red.

Item	Character	Size	Contents
Header	STX	1	STX: 02h
Processing category	'Y'	1	Order index query response command ('Y': 59h)
Separator	','	1	
Number of indexes	Number	2	Number of indexes to be appended afterwards (max. 2 characters)
Separator	','	1	
Sample number 1	Number	13	Max. 13 characters
Separator	','	1	
Patient ID 1	Alphanumeric	13	Max. 13 characters
Separator.	','	1	
Patient name 1	Alphanumeric	13	Max. 13 characters
Separator	','	1	
Animal type 1	Number	2	0 - 99 (max. 2 characters)
Separator	','	1	
Gender 1	Number	1	0: Male 1: Female 9: Unsettled
Separator	','	1	
Age 1	Number	3	999: not set (max. 3 characters)
Separator	','	1	
Number of measurements	Number	2	0 - 5 items = number of measurement items (2 characters fixed (0 padding): '00' ~ '05')
Separator	','	1	
Measurement item 1	English	6	Max. 6 characters
Separator	','	1	
Reference interval range type	Number	1	0 to 6 (1 character fixed)
Separator	','	1	
Measurement item 2	English	6	Max. 6 characters
Separator	','	1	
Reference interval range type	Number	1	0 to 6 (1 character fixed)
Separator	','	1	

Measurement item 3	English	6	Max. 6 characters
Separator	'	1	
Reference interval range type	Number	1	0 to 6 (1 character fixed)
Separator	'	1	
Measurement item 4	English	6	Max. 6 characters
Separator	'	1	
Reference interval range type	Number	1	0 to 6 (1 character fixed)
Separator	'	1	
Measurement item 5	English	6	Max. 6 characters
Separator	'	1	
Reference interval range type	Number	1	0 to 6 (1 character fixed)
Block Separator	ETB	1	
...			
Sample number n	Number	13	Max. 13 characters
Separator	'	1	
Patient ID n	Alphanumeric	13	Max. 13 characters
Separator	'	1	
Patient name n	Alphanumeric	13	Max. 13 characters
Separator	'	1	
Animal type n	Number	2	0 - 99 (max. 2 characters)
Separator	'	1	
Gender n	Number	1	0: male 1: female 9: unspecified
Separator	'	1	
Age n	Number	3	999: not set (max. 3 characters)
Separator	'	1	
Number of measurements	Number	2	0 - 5 items = number of measurement items (2 characters fixed (0 padding): '00' ~ '05')
Separator	'	1	
Measurement item 1	English	6	Max. 6 characters
Separator	'	1	
Reference interval range type	Number	1	0 to 6 (1 character fixed)
Separator	'	1	
Measurement item 2	English	6	Max. 6 characters
Separator	'	1	

Reference interval range type	Number	1	0 to 6 (1 character fixed)
Separator	','	1	
Measurement item 3	English	6	Max. 6 characters
Separator	','	1	
Reference interval range type	Number	1	0 to 6 (1 character fixed)
Separator	','	1	
Measurement item 4	English	6	Max. 6 characters
Separator	','	1	
Reference interval range type	Number	1	0 to 6 (1 character fixed)
Separator	','	1	
Measurement item 5	English	6	Max. 6 characters
Separator	','	1	
Reference interval range type	Number	1	0 to 6 (1 character fixed)
Delimiter	ETX	1	ETX: 03h
BCC			

※The sample number, the number of sheets to be measured and the number of indexes are mandatory.

※If there is no order, the number of indexes shall be set to 0. However, the sample number data added to the query request shall be used for sample number 1.

※If there is an order, the sample number is mandatory and the patient ID and patient name must have data on either of them.

※If the sample number is empty, or if the sample number exists but both patient ID and patient name are empty, the field is invalid.

※The number of measurement sheets indicates the number of measurement items, up to a maximum of five.

※If the number of measurement sheets is 0, it is interpreted as no order.

※If the number of measurement items is large in relation to the number of sheets, the fields after the number of sheets are ignored.

※If the number of measurement items is small in relation to the number of sheets, no error is made and processing continues.

※The maximum number of measurement item fields is five.

※If the number of index fields is large in relation to the number of indexes, the subsequent fields are ignored.

※If the number of index fields is small in relation to the number of indexes, no error is made and processing continues.

※The block delimiter (ETB) in the last index field shall not be added.

7.3 Measurement start notification

Used to notify the start of measurement from the Measuring equipment to an external communication device.

Item	Character	Size	Contents
Header	STX	1	STX: 02h
Processing category	'S'	1	Measurement start notification command ('S': 53h)
Separator	','	1	
Measurement type	English	7	Identification of control or general measurement. CONTROL or NORMAL□ <□Space (20h)>
Separator	','	1	
Date of measurement	Number	10	Date on which the measurement was carried out. (ex : 2006-09-25)
Separator	','	1	
Time of measurement	Number	5	Time at which the measurement was carried out. (ex : 10:50)
Separator	','	1	
Sample number	Number	13	Unique ID for each specimen
Separator	','	1	
Patient ID	Alphanumeric	13	Unique ID for each patient
Separator	','	1	
Patient name	Alphanumeric	13	Patient name
Separator	','	1	
Sample position	Number	2	Position on the disc where the specimen was placed ('01' fixed)
Delimiter	ETX	1	ETX: 03h
BCC			

※Each area shall be of a fixed length and shall be left-justified to store data.

※The sample position shall be fixed at 01.

Example:

When the measurement of the first sample request is started at 10:50 on 12/06/2006

Sample No. 2006061201
Patient ID ABCDEFGHIJKLM
Patient name Taro Fuji

<Transmission data>

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	S	N	N	O	R	M	A	L	□	,	2	0	0	6
2	-	0	6	-	1	2	,	1	0	:	5	0	,	2	0
3	0	6	0	6	1	2	0	1	□	□	□	,	A	B	C
4	D	E	F	G	H	I	J	K	L	M	,	T	a	r	O
5	□	F	u	j	i	□	□	□	□	,	0	1	ETX	BCC	

□ = space

7.4 Measurement result notification

7.4.1 Type 1/2

Used to notify measurement results from the Measuring equipment to an external communication device when set to Type 1/2.

Item	Character	Size	Contents
Header	STX	1	STX: 02h
Processing category	'R'	1	Measurement result notification command ('R': 52h)
Separator	','	1	
Measurement type	English	7	Identification of control or general measurement.
Separator	','	1	
Date of measurement	Number	10	Date on which the measurement was carried out. (ex.2006-09-25)
Separator	','	1	
Time of measurement	Number	5	Time at which the measurement was carried out. (e.g. 10:50)
Separator	','	1	
Sample number	Number	13	Unique ID for each sample
Separator	','	1	
Patient ID	Alphanumeric	13	Unique ID for each patient
Separator	','	1	
Patient name	Alphanumeric	13	Patient name
Separator	','	1	
Animal type	Number	2	0 to 99 (2 characters fixed: 0 padding)
Separator	','	1	
Gender	Number	1	0: Male 1: Female 9: Stand-alone measurement
Separator	','	1	
Age	Number	3	999: Stand-alone measurement (3 characters fixed: 0 padding)
Separator	','	1	
Sample position	Number	2	Position on the disc where the sample was placed

Separator	','	1	
Number of measurements	Number	2	0 - 5 items = number of measurement items
Separator	','	1	
Item name	Alphanumeric	8	Measurement item name (max. 6 digits) + space (20h)
Separator	','	1	
Equal sign		1	Equal or unequal sign
Separator	','	1	
Measurement result	Alphanumeric	9	Measurement result (9 digits)
Unit	Alphanumeric	6	Units (6 digits)
Separator	','	1	
Dilution factor	Number	2	Measured dilution factor (01 -)
Separator	','	1	
Reference interval range lower limit	Number	5	Reference interval range lower limit set value
Separator	','	1	
Reference interval range upper limit	Number	5	Reference interval range upper limit set value
Separator	','	1	
Warning	English Symbols	11	Depends on the following
Delimiter	ETX	1	ETX: 03h
BCC			

Measurement result field

※A measurement result field is added for the measurement result.

※Each area shall be of fixed length and left-justified, with the margins filled with spaces (20h).

※Gender and age fields

In bi-directional communication, the values sent from the host shall be copied as they are.

In the case of stand-alone measurements, '9' shall be embedded in the gender field and '999' in the age field.

※The sample position should be fixed at '01'.

※The number of measurement sheets indicates the number of items to be measured, up to a maximum of five items.

※As there is only one sample type, the 8-digit item name field should be left-justified to contain the name of the measurement item (max. 6 digits) and the margin should be filled with spaces (20h).

※For information on warning areas, see 7.4.5 Warning specifications.

7.4.2 Type 3

Used to notify the measurement result from the Measuring equipment to an external communication device when Type 3 is set.

Item	Character	Size	Contents
Header	STX	1	STX: 02h
Measurement type	English	7	Identification of control or general measurement.
Date of measurement	Number	10	Date when the measurement was carried out. (ex. 2006-09-25)
Time of measurement	Number	5	Time at which the measurement was carried out. (ex. 10:50)
Sequence No.	Number	13	Unique number for each sample
Sample ID	Alphanumeric	13	Unique ID for each sample
Sample position	Number	2	Position on the disc where the sample was placed ('01' fixed)
Item name / Sample type	Alphanumeric	7	Measurement item name (max. 6 digits) + space (20h)
Equal sign	Alphanumeric	1	Equal or unequal sign
Measurement result	Number	9	Measurement result (9 digits)
Unit	English	6	Units (6 digits)
Dilution factor	Number	2	Measurement dilution factor (01 -)
Warning	English Symbols	11	According to the following
Delimiter	ETX	1	ETX: 03h

Measurement result field

※Add the measurement result field for the measurement result.

※For warning areas, see 7.4.5 Warning specifications.

7.4.3 Type 1 + reference interval range / Type 2 + reference interval range

Used when the measurement result is notified from the analyzer to an external communication device when it is set to Type 1 + reference interval range / Type 2 + reference interval range.

Item	Character	Size	Contents
Header	STX	1	STX: 02h
Processing category	'T'	1	Measurement result notification command ('T': 54h)
Separator	','	1	
Measurement type	English	7	Identification of control or general measurement. CONTROL or NORMAL
Separator	','	1	
Date of measurement	Number	10	Date on which the measurement was carried out. (ex. 2006-09-25)
Separator	','	1	
Time of measurement	Number	5	Time at which the measurement was carried out. (ex. 10:50)
Separator	','	1	
Sample number	Number	13	Unique ID for each sample
Separator	','	1	
Patient ID	Alphanumeric	13	Unique ID for each patient
Separator	','	1	
Patient name	Alphanumeric	13	Patient name
Separator	','	1	
Animal type	Number	2	0-99 (2 characters fixed: 0 padding)
Separator	','	1	
Gender	Number	1	0: Male 1: Female 9: Stand-alone measurement
Separator	','	1	
Age	Number	3	999: Stand-alone measurement (3 characters fixed: 0 padding)
Separator	','	1	
Sample position	Number	2	Position on the disc where the sample was placed ('01' fixed)
Separator	','	1	
Number of measurements	Number	2	0-5 items = number of measurement items (2 characters fixed (0 padding): '00'~ '05')

Separator	,	1	
Item name	Alphanumeric	8	Measurement item name (max. 6 digits) + space (20h)
Separator	,	1	
Equal sign		1	Equal or unequal sign
Separator	,	1	
Measurement result	Alphanumeric	9	Measurement result (9 digits)
Unit	Alphanumeric	6	Units (6 digits)
Separator	,	1	
Dilution factor	Number	2	Measured dilution factor (01 -)
Separator	,	1	
Reference interval range lower limit	Number	5	Reference interval range lower limit set value
Separator	,	1	
Reference interval range upper limit	Number	5	Reference interval range upper limit set value
Separator	,	1	
Warning	English Symbols	11	According to the following
Item Separator	,	1	Delimited by
Reference interval range type	Number	1	0 to 6 (1 character fixed)
Delimiter	ETX	1	ETX: 03h
BCC			

Measurement result field

※A measurement result field is added for the measurement result.

※Each area shall be of fixed length and left-justified, with the margins filled with spaces (20h).

※Gender and age fields

In bi-directional communication, the values sent from the host shall be copied as they are.

In the case of stand-alone measurements, '9' shall be embedded in the gender field and '999' in the age field.

※The sample position should be fixed at '01'.

※The number of measurement sheets indicates the number of items to be measured, up to a maximum of five items.

※As there is only one sample type, the 8-digit item name field should be left-justified to contain the name of the measurement item (maximum 6 digits) and the blank space should be filled with spaces (20h).

※For warning areas, see 7.4.5 Warning specifications.

7.4.4 Type 3 + reference interval range

Used to notify the measurement result from the Measuring equipment to an external communication device when the Type 3 + reference interval range is set.

Item	Character	Size	Contents
Header	STX	1	STX: 02h
Measurement type	English	7	Identification of control or general measurement. CONTROL or NORMAL
Date of measurement	Number	10	Date when the measurement was carried out. (ex. 2006-09-25)
Time of measurement	Number	5	Time at which the measurement was carried out. (ex. 10:50)
Sequence No.	Number	13	Unique number for each sample
Sample ID	Alphanumeric	13	Unique ID for each sample
Sample position	Number	2	Position on the disc where the sample was placed ('01' fixed)
Item name / Sample type	Alphanumeric	7	Measured item name (max. 6 digits) + space (20h)
Equal sign	Alphanumeric	1	Equal or unequal sign
Measurement result	Number	9	Measurement result (9 digits)
Unit	English	6	Unit (6 digits)
Dilution factor	Number	2	Measured dilution factor (01 ~)
Warning	English Symbols	11	According to the following
Animal type	Number	2	0 to 99 (2 characters fixed: 0 padding)
Reference interval range type	Number	1	0 - 6 (1 character fixed)
Reference interval range lower limit	Number	5	Reference interval range lower limit set value
Reference interval range upper limit	Number	5	Reference interval range upper limit set value
Delimiter	ETX	1	ETX: 03h

Measurement result field

※Add the measurement result field for the measurement result.

※For information on warning areas, see 7.4.5 Warning specifications.

7.4.5 Warning specifications

The warnings area assignments are shown below.

If no error occurs in each assignment, a space (20h) shall be used.

Position (feed order)	Sending character	Explanation
1st character	H	The measurement value is greater than the upper limit of the set reference interval range.
	L	The measurement value is less than the lower limit of the set reference interval range.
2nd character	@	The measured value is outside the quantitative range.
3rd character	#	Reagent chip expired
4th character	+	Abnormal temperature during or after measurement (high temperature side)
	—	Abnormal temperature during and after measurement (low temperature side)
5th character	※	Waste box opened during measurement
6th character		Always space (20h)
7th character		Always space (20h)
8th character		Always space (20h)
9th character		Always space (20h)
10th character		Always space (20h)
11th character		Always space (20h)

Transmission example: Type 1 / Type 2

Measurement type	NORMAL: Normal measurement
Date of measurement	2009-07-13
Time of measurement	19:12
Sample number	2009071301
Patient ID	ABCDEFGF
Patient Name	Taro Fuji (not shown on LCD display)
Animal type	15: Cat
Gender	0: Male
Age	003: 3 years old
Sample position	01 (Fixed)
Number of measurements	01: 1 item
Item name	v-TSH
Equal sign	=
Measurement result	250.6
Unit of measurement	mg/L
Dilution factor	01
Reference interval range lower limit	111
Reference interval range upper limit	222
Warning	☐ @# ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

<Transmission data>

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	R	,	N	O	R	M	A	L	☐	,	2	0	0	9
2	-	0	7	-	1	3	,	1	9	:	1	2	,	2	0
3	0	9	0	7	1	3	0	1	☐	☐	☐	,	A	B	C
4	D	E	F	G	☐	☐	☐	☐	☐	☐	,	T	a	r	o
5	☐	F	u	j	i	☐	☐	☐	☐	,	1	5	,	0	,
6	0	0	3	,	0	1	,	0	1	,	v	-	T	S	H
7	☐	☐	☐	,	=	,	2	5	0	.	6	☐	☐	☐	☐
8	m	g	/	L	☐	☐	,	0	1	,	1	1	1	☐	☐
9	,	2	2	2	☐	☐	,	☐	@	#	☐	☐	☐	☐	☐
10	☐	☐	☐	ETX	BCC										

☐ = space

Transmission example: Type 3

Measurement type NORMAL: Normal measurement
Date of measurement 2009-07-13
Time of measurement 19:12
Sequence number 2009071301
Sample ID ABCDEFG
Sample position 01 (Fixed)
Item name v-TSH
Equal sign =
Measurement result 250.6
Unit of measurement mg/L
Dilution factor 01
Warning □@#□□□□□□□□
Animal type 15: Cat
Reference interval range type 1
Reference interval range lower limit 111
Reference interval range upper limit 222

<Transmission data>

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	N	O	R	M	A	L	□	2	0	0	9	-	0	7
2	-	1	3	1	9	:	1	2	2	0	0	9	0	7	1
3	3	0	1	□	□	□	A	B	C	D	E	F	G	□	□
4	□	□	□	□	0	1	v	-	T	S	H	□	□	=	2
5	5	0	.	6	□	□	□	□	m	g	/	L	□	□	0
6	1	□	@	#	□	□	□	□	□	□	□	□	1	5	1
7	1	1	1	□	□	2	2	2	□	□	ETX				

□ = space

7.5 Error information details

Used to inform the external communication device from the analyzer of error information generated by the analyzer.

Item	Character	Size	Contents
Header	STX	1	STX: 02h
Processing category	'E'	1	Error information notification command
Separator	','	1	
Date of error occurrence	Number	10	Date on which the error occurred. (ex. 2006-06-12)
Separator	','	1	
Time of error occurrence	Number	8	Time at which the measurement was carried out. (ex. 10:30:50)
Separator	','	1	
Error number	Alphanumeric	5	Error number
Separator	','	1	
Number of additional information	Number	1	Number of additional information (0-9)
Separator	','	1	
Additional information 1	Alphanumeric	6	Error additional information
Separator	','	1	
...			
Additional information n	Alphanumeric	6	Error additional information
Delimiter	ETX	1	ETX: 03h
BCC			

Additional information field

※Additional information fields are added for the number of additional information fields.

※The delimiter of the last additional information field shall not be added.

※Each area shall be of fixed length and data shall be left-justified and the margins shall be filled with spaces (20h).

Example: When suction blockage error E0102 occurs

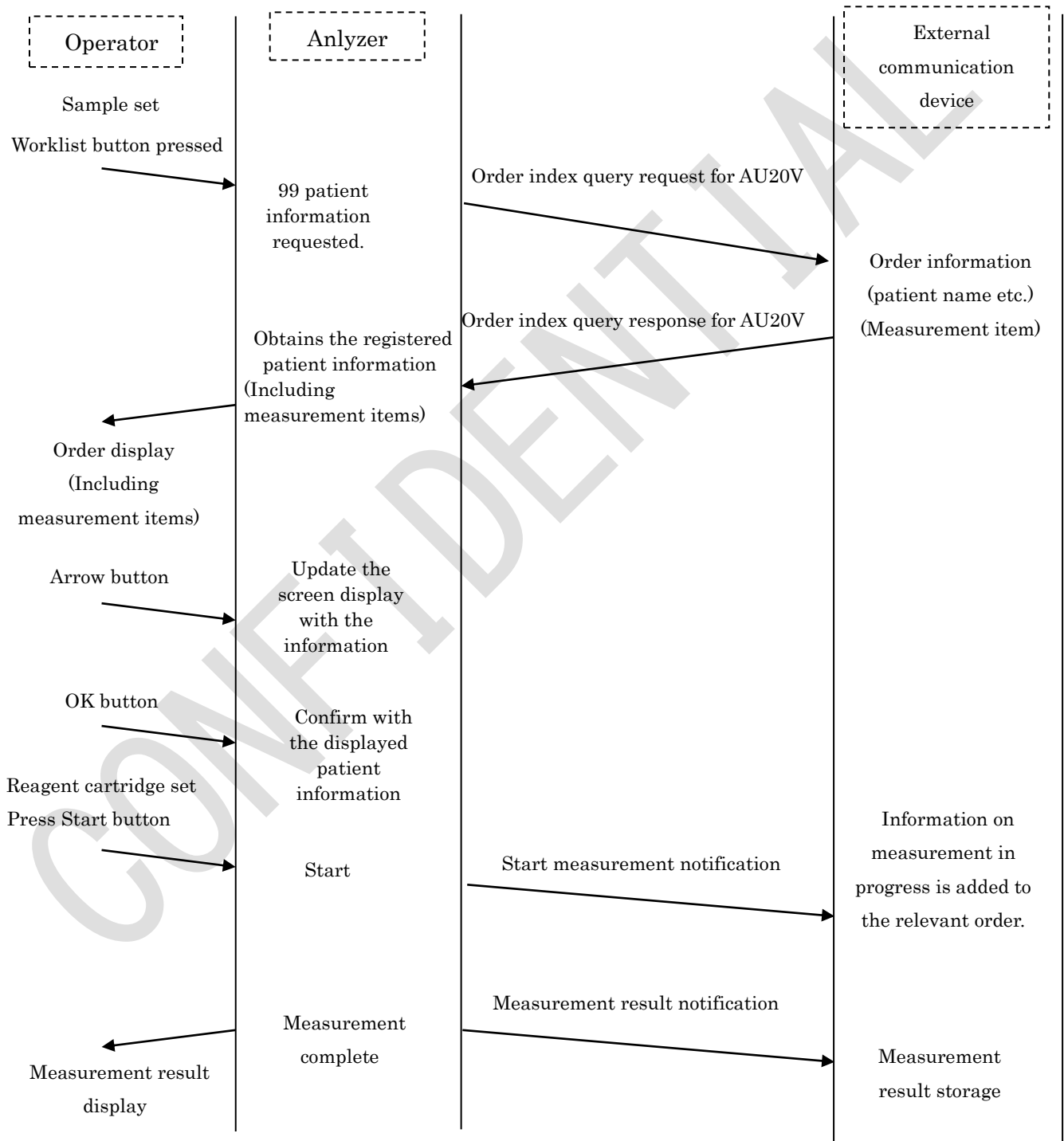
<Transmission data>

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	STX	E	,	2	0	0	6	-	0	6	-	1	2	,	1
2	0	:	3	0	:	5	0	,	E	0	1	0	2	,	1
3	,	1	.	0	0	0	□	ETX	BCC						

□= Space

8 Transmission/operating flow

8.1 1 sample request measurement (Type 1 / Type 1 + reference interval range)



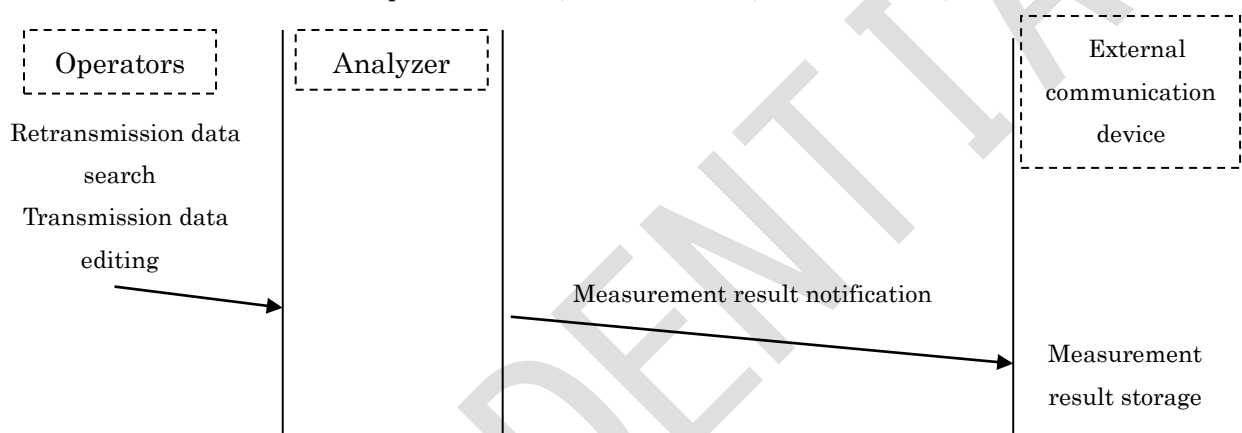
※By pressing the Worklist button, the index (including measurement items) is queried (for 99 persons).

※By pressing the arrow button, only the display of the acquired information is switched.

※Pressing the Start button starts the measurement of the retained samples.

8.2 Measurement data retransmission (Type 1,2 / Type 1,2 + reference interval range)

Retransmission of measurement data in the event of non-connection of an external communication device (power failure, no connection, disconnection, etc.).



8.3 Measurement data transmission (type 1,2,3 / type 1,2,3 + reference interval range)

Transmission of measurement data in streaming communication.

